Laplace transforms for higher order equations

As we continue with this theme of expanding what we learned about second order equations to third and higher order equations, we can of course consider the Laplace transform.

For instance, let's add the Laplace transform of the third derivative to our list of formulas.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(f'''(t)) = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$$

We should start to see a pattern here for the Laplace transform of higher order derivatives, and we could add even more transforms.

$$\mathscr{L}(f^{(4)}(t)) = s^4F(s) - s^3f(0) - s^2f'(0) - sf''(0) - f'''(0)$$

$$\mathscr{L}(f^{(5)}(t)) = s^5F(s) - s^4f(0) - s^3f'(0) - s^2f''(0) - sf'''(0) - f^{(4)}(0)$$

...

Which means we can now apply these to find the Laplace transform of a higher order differential equation. Let's do an example, using the same differential equation we've been working with through the last few lessons.

Example

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Use a Laplace transform to find the solution to the third order differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$.

$$y''' - 6y'' + 12y' - 8y = 3e^{2t} + 4e^{3t}$$

Apply the transform to both sides of the differential equation.

$$\mathcal{L}(y''') - 6\mathcal{L}(y'') + 12\mathcal{L}(y') - 8\mathcal{L}(y) = 3\mathcal{L}(e^{2t}) + 4\mathcal{L}(e^{3t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - 6(s^2Y(s) - sy(0) - y'(0))$$

$$+12(sY(s) - y(0)) - 8Y(s) = 3\left(\frac{1}{s-2}\right) + 4\left(\frac{1}{s-3}\right)$$

$$s^3Y(s) - 6s^2Y(s) + 12sY(s) - 8Y(s) = 3\left(\frac{1}{s-2}\right) + 4\left(\frac{1}{s-3}\right)$$

Solve for $Y(s)$.

$$(s^3 - 6s^2 + 12s - 8)Y(s) = 3\left(\frac{1}{s-2}\right) + 4\left(\frac{1}{s-3}\right)$$

$$Y(s) = \frac{3}{(s-2)(s^3 - 6s^2 + 12s - 8)} + \frac{4}{(s-3)(s^3 - 6s^2 + 12s - 8)}$$

$$Y(s) = \frac{3}{(s-2)^4} + \frac{4}{(s-2)^3(s-3)}$$

Combine fractions.

$$Y(s) = \frac{3(s-3)}{(s-2)^4(s-3)} + \frac{4(s-2)}{(s-2)^4(s-3)}$$

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$$Y(s) = \frac{7s - 17}{(s-2)^4(s-3)}$$

Apply a partial fractions decompositions.

$$\frac{7s - 17}{(s-2)^4(s-3)} = \frac{A}{s-2} + \frac{B}{(s-2)^2} + \frac{C}{(s-2)^3} + \frac{D}{(s-2)^4} + \frac{E}{s-3}$$

We find $A = -4$, $B = -4$, $C = -4$, $D = 3$, and $E = 4$, so the Laplace transform becomes

$$Y(s) = \frac{-4}{s-2} + \frac{-4}{(s-2)^2} + \frac{-4}{(s-2)^3} + \frac{3}{(s-2)^4} + \frac{4}{s-3}$$

$$Y(s) = -4\left(\frac{1}{s-2}\right) - 4\left(\frac{1}{(s-2)^2}\right) - 4\left(\frac{1}{(s-2)^3}\right) + 3\left(\frac{1}{(s-2)^4}\right) + 4\left(\frac{1}{s-3}\right)$$

$$Y(s) = -4\left(\frac{1}{s-2}\right) - 4\left(\frac{1}{s-2}\right)^2 - 4\left(\frac{1}{s-2}\right)^3 + 3\left(\frac{1}{s-2}\right)^4 + 4\left(\frac{1}{s-3}\right)$$

We can easily apply the inverse transform to the first and last terms,

$$y(t) = -4e^{2t} - 4\left(\frac{1}{s-2}\right)^2 - 4\left(\frac{1}{s-2}\right)^3 + 3\left(\frac{1}{s-2}\right)^4 + 4e^{3t}$$

And as we know from our table of Laplace transforms,

$$\mathscr{L}(t^n e^{at}) = \frac{n!}{(s-a)^{n+1}}$$

so we'll use this formula with $n = 1$ to apply the inverse transform to the second term.

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$$y(t) = -4e^{2t} - 4te^{2t} - 4\left(\frac{1}{(s-2)^3}\right) + 3\left(\frac{1}{(s-2)^4}\right) + 4e^{3t}$$

Using $n = 2$ to take the inverse transform of the third term, we get

$$y(t) = -4e^{2t} - 4te^{2t} - \frac{4}{2!}\left(\frac{2!}{(s-2)^3}\right) + 3\left(\frac{1}{(s-2)^4}\right) + 4e^{3t}$$

$$y(t) = -4e^{2t} - 4te^{2t} - \frac{4}{2!}t^2e^{2t} + 3\left(\frac{1}{(s-2)^4}\right) + 4e^{3t}$$

Using $n = 3$ to take the inverse transform of the fourth term, we get

$$y(t) = -4e^{2t} - 4te^{2t} - \frac{4}{2!}t^2e^{2t} + \frac{3}{3!}\left(\frac{3!}{(s-2)^4}\right) + 4e^{3t}$$

$$y(t) = -4e^{2t} - 4te^{2t} - \frac{4}{2!}t^2e^{2t} + \frac{3}{3!}t^3e^{2t} + 4e^{3t}$$

So the general solution to the differential equation is

$$y(t) = 4e^{3t} - 4e^{2t} - 4te^{2t} - \frac{4}{2!}t^2e^{2t} + \frac{3}{3!}t^3e^{2t}$$

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